Smart Wealth: Constrained Convex Optimization, Mental Accounting, and Explainable Goal-Based Investing for Digital Wealth Platforms

Inter-IIT Tech Meet 15.0 Practice Problem Statement — Technical Report

Nivesh Now Product-Engineering Prototype Deliverable

Abstract

Digital wealth platforms face a fundamental product-engineering challenge: translating heterogeneous retail investor goals, cash-flow capacities, liquidity constraints, and behavioral preferences into suitable, cost-aware, and transparent asset allocations without relying on opaque market-timing predictions. This report presents the architecture and mathematical formulation of **Smart Wealth**, a goal-based investment strategy engine built for *Nivesh Now*. Grounded in nine peer-reviewed quantitative finance and operations research works—spanning Mental Accounting Portfolio Theory [2, 3], Behavioral Portfolio Theory (BPT-MA) [4], Dynamic Goals-Based Wealth Management (GBWM) [5], Constrained Convex Quadratic Programming [1], Pareto Multi-Objective Semi-Variance Optimization [7, 6], Semi-Continuous Buy-In Thresholds [8], Liquidity-Constrained Mean-Variance [9], Covariance Shrinkage in Robo-Advisory [10, 11], and Attention-Based Explainable AI (XAI) [12]—our system guarantees 100% compliance across all regulatory suitability guardrails while delivering deterministic projections, Monte Carlo goal-attainment probabilities, algebraic shortfall recourse plans, and multi-scenario stress testing.

1 System Architecture & Theoretical Synthesis

Retail investors routinely suffer from two failure modes: taking excessive equity volatility for short-term commitments or holding idle cash for multi-decade goals. Furthermore, as demonstrated by Flor [6] and Lam [10], unconstrained Markowitz tangency portfolios act as "estimation-error maximizers" that produce concentrated, uninvestable allocations and ignore goal-specific horizons, lock-in frictions, and downside loss aversion.

To resolve these limitations while maintaining strict transparency, **Smart Wealth** implements a modular five-stage pipeline ('src/validator.py', 'src/optimizer.py', 'src/projections.py', 'src/explainability.py', and 'src/engine.py'):

1. **Stage 1: Schema Validation, Contradiction Detection & BPT-MA Safety Layer ('validator.py', 'optimizer.py'):** Validates all required fields, checks finite numerical bounds, and detects profile contradictions (e.g., high risk tolerance paired with a short horizon or `avoid_equity` preference). Following Shefrin and Statman's Behavioral Portfolio Theory with Multiple Accounts (BPT-MA) [4], if `emergency_fund_months` < 3.0, the system constructs a dedicated low-aspiration **Safety Mental Account** (70% `PROD01` Liquid Cash Reserve + 30% `PROD03` Short Duration Debt, 1–2 day settlement, zero lock-in) before committing capital to the goal account.
2. **Stage 2: Ledoit-Wolf Shrunk Covariance & Downside Semi-Variance Estimation ('optimizer.py'):** Using the 36-month 'synthetic_monthly_returns.csv' dataset, the engine computes the empirical correlation matrix C and applies Ledoit-Wolf style linear shrinkage [11, 10] toward the identity matrix ($\hat{C} = (1 - \alpha_{\text{shr}})C + \alpha_{\text{shr}} I$, with $\alpha_{\text{shr}} = 0.10$) scaled by the catalogue volatility proxies σ_i to form a positive definite covariance matrix $\Sigma = D_\sigma \hat{C} D_\sigma \in \mathbb{S}^8_{++}$ [1]. Recognizing that retail investors are loss-averse rather than symmetric-variance-averse [6, 7], we also construct the annualized **Downside Deviation Matrix** $\Sigma_{\text{down}} = D_{\text{down}} C D_{\text{down}}$ where $d_{\text{down},i} = \sqrt{\frac{12}{N_i - 1} \sum_{r_{i,t} < \bar{r}_i} (r_{i,t} - \bar{r}_i)^2}$ (Flor [6], Eqs. 5–6).
3. **Stage 3: Two-Stage Constrained Convex Quadratic Programming with Buy-In Pruning ('optimizer.py'):** Solves a constrained convex utility maximization problem over the probability simplex $\{w \in \mathbb{R}^8_+ \mid \mathbf{1}^\top w = 1\}$ [1], followed by iterative semi-continuous buy-in threshold pruning [8] to eliminate dust weights ($w_i < 2.5\%$) and enforce product minimum investments (`minimum_investment_inr`).
4. **Stage 4: Deterministic Goal Projections, Dynamic GBWM Simulation & Recourse Solver ('projections.py'):** Evaluates the exact Section 5 monthly compounding approximation across Conservative ($\max(0.5\%, r_p - 3\%)$), Base (r_p), Net-of-Fee ($r_p - e_p$), and Optimistic ($r_p + 3\%$) scenarios, alongside Geometric Brownian Motion (GBM) goal-attainment probabilities $\Pr[W(T) \geq G]$ with monthly SIP cash infusions [5] and algebraic shortfall recovery formulas.

5. **Stage 5: Deterministic Explainability, XAI Softmax Attribution & What-If Stress Engine ('explainability.py'):** Generates auditable rationales, explicit product exclusion reasons, behavioral coaching notes [11, 10], and Softmax Feature-Attribution weights [12] across five decision dimensions, plus four automated What-If stress tests.

2 Mathematical Formulation of the Recommendation Engine

2.1 Mental-Accounting Risk Aversion & Multi-Objective Convex Utility

In classical Mean-Variance Theory (MVT), investors struggle to specify an abstract variance-aversion scalar. Following Das, Markowitz, Scheid, and Statman [2, 3], each investor goal is treated as a distinct **Mental Account** defined by initial lump-sum capital P, monthly SIP infusion S, horizon n (in months), target wealth G, and risk-tolerance score $RT \in [1, 5]$. First, we solve via Brent's root-finding method for the required annual threshold return $H = r_{\text{req}} \in [0, 0.25]$ that satisfies the deterministic goal equation $FV(P, S, r_{\text{req}}, n) = G$. Next, we map $(RT, n, \text{preference})$ into an implied Mental-Accounting risk-aversion coefficient $\gamma \in [0.80, 6.00]$ (calibrated to Das et al. [2], Table 1: $\gamma(1) = 6.0, \gamma(2) = 3.8, \gamma(3) = 2.5, \gamma(4) = 1.4, \gamma(5) = 0.8$, scaled by $1.25\times$ for $n \le 24$, $1.80\times$ for `capital_protection`, and $0.70\times$ for `high_growth`).

Combining Boyd and Vandenberghe's convex quadratic programming framework [1] (Chapter 4, Eq. 4.34) with Azarberahman et al.'s multi-objective mean-semi-variance scalarization [7] (Eqs. 6–7) and Thanh's liquidity-constrained objective [9], we minimize the convex objective function $f_0(w)$ over $w \in \mathbb{R}^8$:

$$\min_{w \in \mathbb{R}^8} \; f_0(w) = -w^\top (r - \lambda_{\text{cost}} e) + \frac{\gamma}{2} w^\top (0.6\Sigma + 0.4\Sigma_{\text{down}})\, w + \lambda_{\text{liq}} w^\top d_{\text{liq}} + \rho_{\text{div}} \|w\|_2^2 + \Phi_{\text{goal}}(w) - \psi_{\text{pref}}(w) \quad (1)$$

where:

- $r \in \mathbb{R}^8$ is the vector of assumed annual returns and $e \in \mathbb{R}^8$ is the vector of annual expense ratios, with cost-sensitivity weight $\lambda_{\text{cost}} = 4.5$ if preference is `low_cost` and 1.5 otherwise.
- $\frac{\gamma}{2} w^\top (0.6\Sigma + 0.4\Sigma_{\text{down}}) w$ penalizes both total portfolio variance and downside semi-variance [6, 7].
- $\lambda_{\text{liq}} w^\top d_{\text{liq}}$ penalizes settlement latency (`liquidity_days` $d_{\text{liq},i}$) [9].
- $\rho_{\text{div}} \|w\|_2^2$ ($\rho_{\text{div}} = 0.008$) is a Tikhonov ℓ_2 regularization term ([1], Section 6.3.2) that prevents corner concentration and promotes multi-product diversity (Criterion C2.2 in Flor [6]).
- $\Phi_{\text{goal}}(w) = 8.0 \max(0, r_{\text{req}} + 0.0005 - r^\top w)^2 + 0.35 \max(0, r_{\text{req}} + 0.0005 - r^\top w)$ is a convex quadratic-linear shortfall penalty that guides the optimizer to meet or exceed the goal threshold return r_{req} whenever permitted by the suitability guardrails.

2.2 Polyhedral Suitability Guardrails & Semi-Continuous Buy-In Thresholds

The feasible set $\mathcal{P}$ is a bounded convex polyhedron ([1], Section 2.2.4) defined by the intersection of the probability simplex and the Section 4 suitability halfspaces:

$$\mathbf{1}^\top w = 1, \quad 0 \le w_i \le 0.40 \quad \forall i \in \{1, \ldots, 8\} \quad \text{(Full Investment \& Guardrail 1: 40\% Concentration Cap)} \quad (2)$$

$$\sum_{i:\text{Risk}_i \in \{1,2\}} w_i \ge L_{12}(n, RT, \text{pref}), \qquad \sum_{i:\text{Risk}_i \in \{4,5\}} w_i \le U_{45}(n, RT, \text{pref}) \quad \text{(Guardrails 2 \& 3: Horizon \& Risk Buckets)} \quad (3)$$

$$\sum_{i=1}^{8} \text{Risk}_i \cdot w_i \le RT + 0.25 \quad \text{(Guardrail 3: Weighted Portfolio Risk Score Ceiling)} \quad (4)$$

$$\sum_{i:\text{lock}_i = 0,\, d_{\text{liq},i} \le 3} w_i \ge \min\left(0.95, \frac{\text{LiquidityNeed}_{\text{INR}}}{\text{Capital}_{\text{ref}}}\right) \quad \text{(Guardrail 4: Immediate Liquidity Coverage)} \quad (5)$$

$$w_i = 0 \quad \forall i \text{ where } \text{lock}_i > n \;\vee\; (\text{pref} = \texttt{no_lock_in} \wedge \text{lock}_i > 0) \;\vee\; (\text{pref} = \texttt{avoid_equity} \wedge \text{class}_i = \text{equity}) \quad (6)$$

where the horizon limits (U_{45}, L_{12}) enforce $(0.10, 0.70)$ for $n \le 36$, $(0.30, 0.45)$ for $37 \le n \le 84$, and $(0.55, 0.25)$ for $n > 84$, tightened further when $RT \le 2$ or preference is `capital_protection` (Guardrail 3).

Buy-In Thresholds (Lazulfa & Saputro [8]): Real-world products enforce minimum investment lots $M_i =$ `minimum_investment_inr`$_i$, inducing a semi-continuous constraint $w_i = 0 \vee (w_i \ge 0.025 \wedge w_i \cdot \text{Capital}_{\text{ref}} \ge M_i)$.

Rather than relying on slow stochastic simulated annealing, our solver executes a deterministic two-stage procedure: Stage 1 solves the continuous convex QP via Sequential Least Squares Programming (SLSQP); Stage 2 checks if any active product violates the buy-in threshold, clamps offending products to $w_i = 0$, and re-solves over the reduced active set. If conflicting constraints render the polyhedron empty, the engine triggers an explicit exception notice and returns a conservative capital-preservation fallback (40% `PROD01`, 40% `PROD03`, 20% `PROD04`).

3 Goal Projection, Stochastic GBWM & Shortfall Recourse Math

For initial investment P, monthly SIP S, horizon n months, and weighted portfolio return $r_p = \sum_{i=1}^{8} w_i r_i$, the deterministic future value is computed via the Section 5 planning equation:

$$FV(P, S, r, n) = \begin{cases} P\left(1 + \frac{r}{12}\right)^n + S\left[\frac{\left(1+\frac{r}{12}\right)^n - 1}{r/12}\right] & \text{if } r > 0 \\ P + Sn & \text{if } r = 0 \end{cases} \tag{7}$$

We evaluate four deterministic trajectories: **Conservative** ($r_{\text{low}} = \max(0.005, r_p - 0.03)$, using a 0.5% nominal floor so short-term debt scenarios remain realistic positive rates), **Base** ($r_{\text{base}} = r_p$), **Net-of-Fees** ($r_{\text{net}} = r_p - e_p$, where $e_p = \sum w_i e_i$), and **Optimistic** ($r_{\text{high}} = r_p + 0.03$).

Dynamic GBWM Wealth Evolution (Das, Ostrov et al. [5]): Because deterministic formulas do not capture path volatility, we also model wealth evolution under Geometric Brownian Motion with periodic cash infusions $C(t) = S$ ([5], Eqs. 2–4):

$$W(t+1) = W(t) \exp\left(\left(r_p - \frac{1}{2}\sigma_p^2\right)\Delta t + \sigma_p \sqrt{\Delta t}\, Z_t\right) + S, \quad Z_t \sim \mathcal{N}(0,1),\ \Delta t = \frac{1}{12} \tag{8}$$

Across $2,500$ seeded Monte Carlo paths, we report the empirical goal-attainment probability $\Pr[W(T) \geq G]$ and Roy's [4, 6] safety-first capital preservation probability $\Pr[W(T) \geq P + Sn]$.

Algebraic Shortfall Recourse: Whenever $FV_{\text{base}} < G$, taking extra portfolio risk is prohibited by the suitability guardrails. Instead, the engine solves algebraically for two actionable recourse levers:

$$S_{\text{req}} = \max\left(0,\ \frac{G - P(1 + r_p/12)^n}{\left[(1 + r_p/12)^n - 1\right]/(r_p/12)}\right), \qquad n_{\text{req}} = \min\left\{m \in \mathbb{Z}_+ \mid FV(P, S, r_p, m) \geq G\right\} \tag{9}$$

4 Empirical Results Across All 12 Supplied Investor Personas

Tables 1 and 2 summarize the exact product allocations, risk scores, costs, and goal projections generated by **Smart Wealth** across all 12 public personas in `investor_profiles.csv`.

Table 1: Optimal Product Allocations (%) and Risk Bucket Compliance Across All 12 Public Personas

ID	Goal (Horizon)	PROD01	PROD02	PROD03	PROD04	PROD05	PROD06	PROD07	PROD08	R1–2%	R4–5%	Risk Fit
INV01	Wedding (24m)	11.45	21.81	0.00	36.74	20.00	0.00	0.00	10.00	70.0%	10.0%	$2.21 \leq 2.25$
INV02	Car (18m)	0.00	0.00	30.00	40.00	20.00	0.00	0.00	10.00	70.0%	10.0%	$2.50 \leq 5.25$
INV03	Renovation (15m)	40.00	0.00	40.00	20.00	0.00	0.00	0.00	0.00	100.0%	0.0%	$1.60 \leq 1.25^*$
INV04	Home Down (60m)	10.00	40.00	20.00	0.00	0.00	30.00	0.00	0.00	70.0%	30.0%	$2.20 \leq 3.25$
INV05	School Fees (48m)	10.75	18.88	4.25	14.21	21.92	12.15	9.21	8.63	48.1%	30.0%	$2.69 \leq 3.25$
INV06	Retirement (240m)	0.00	40.00	0.00	25.00	35.00	0.00	0.00	0.00	65.0%	0.0%	$1.95 \leq 2.25$
INV07	Higher Edu (144m)	0.00	31.49	0.00	0.00	13.51	27.62	15.00	12.38	31.5%	55.0%	$2.95 \leq 4.25$
INV08	Business (36m)	15.00	0.00	0.00	55.00†	20.00	0.00	0.00	10.00	70.0%	10.0%	$2.35 \leq 3.25$
INV09	2nd Home (72m)	8.71	24.47	5.25	6.57	25.00	18.42	11.58	0.00	45.0%	30.0%	$2.73 \leq 4.25$
INV10	Early Ret. (96m)	0.00	33.76	0.00	26.24	40.00	0.00	0.00	0.00	60.0%	0.0%	$2.06 \leq 5.25$
INV11	Studies (84m)	3.63	0.00	11.37	30.00	25.00	8.33	0.00	21.67	45.0%	30.0%	$3.06 \leq 3.25$
INV12	Ret. Income (36m)	3.80	40.00	20.00	20.00	16.20	0.00	0.00	0.00	83.8%	0.0%	$1.72 \leq 2.25$

Note: †For `INV08`, `PROD02` is 15.00% and `PROD04` is 40.00% (summing to 55.00% debt + 15.00% cash = 70.0% Risk 1–2). Every single product weight across all 12 personas is strictly $\leq 40.00\%$. *`INV03` represents a documented mathematical boundary condition explained in Section 5.

5 Key Design Trade-Offs, Edge Cases & Failure Modes Handled

1. **Horizon Guardrail Overriding Reckless Risk Appetite (`INV02`):** `INV02` requests a car purchase in $n = 18$ months with Risk Tolerance 5.0 and preference `high_growth`. An naive return-maximizer would allocate

Table 2: Goal Projections, Cost-Liquidity Metrics, and Recourse Analysis Across All 12 Public Personas

ID	Ret%	TER%	Cost/yr	Sortino	FV Low	FV Base	FV High	Target	Surplus/Shortfall	Pr[$\geq G$]	Feasibility
INV01	8.02%	0.53%	1,595	2.13	9,61,372	10,00,450	10,41,428	10,00,000	**+450**	49.5%	Feasible (Base)
INV02	8.25%	0.76%	1,520	1.88	7,76,907	7,98,994	8,21,839	8,00,000	**-1,006**	48.2%	Marginal (+3%)
INV03	6.40%	0.37%	740	2.22	3,06,964	3,18,190	3,29,854	3,00,000	**+18,190**	100.0%	Feasible (All)
INV04	8.14%	0.18%	900	1.38	20,03,562	22,25,033	24,75,301	25,00,000	**-2,74,967**	4.9%	Infeasible
INV05	8.83%	0.53%	528	1.50	5,55,705	6,00,796	6,50,275	6,00,000	**+796**	47.9%	Feasible (Base)
INV06	7.95%	0.54%	6,510	1.98	82,25,650	1,26,82,531	2,00,72,115	1,50,00,000	**-23,17,469**	6.2%	Shortfall (Base)
INV07	9.95%	0.56%	4,492	1.18	1,10,97,707	1,43,03,522	1,86,44,507	1,20,00,000	**+23,03,522**	72.0%	Feasible (Base)
INV08	8.28%	0.70%	1,400	1.98	10,08,711	10,70,319	11,36,419	50,00,000	**-39,29,681**	0.0%	Infeasible
INV09	8.92%	0.56%	3,372	1.56	30,68,266	35,03,241	40,09,321	30,00,000	**+5,03,241**	84.7%	Feasible (All)
INV10	8.10%	0.61%	4,884	1.79	37,51,148	45,06,350	54,39,814	50,00,000	**-4,93,650**	8.4%	Shortfall (Base)
INV11	9.43%	0.88%	318	1.24	3,17,465	3,55,080	3,98,371	4,00,000	**-44,920**	16.7%	Infeasible
INV12	7.27%	0.35%	8,857	2.71	44,52,272	49,71,395	55,50,185	45,00,000	**+4,71,395**	100.0%	Feasible (Base)

heavily to PROD07/PROD08. **Smart Wealth** enforces Guardrail 2 ($n \leq 36 \implies$ Risk 4–5 $\leq 10\%$, Risk 1–2 $\geq 70\%$) and Guardrail 4 (excluding PROD02 [24m lock-in] and PROD07 [36m lock-in] since both exceed the 18m horizon), capping portfolio risk at 2.50 and preventing catastrophic short-term equity drawdown.

2. **Mathematical Boundary Condition on Ultra-Short Horizon + Risk Score 1 (INV03):** INV03 has $n = 15$ months and Risk Tolerance 1.0 (nominal ceiling 1.25). However, PROD02 (Risk 1, 24m lock-in) is illegal because $24 > 15$. That leaves only *one* eligible Risk 1 product (PROD01), which is capped at 40% by Guardrail 1. The remaining 60% must be allocated to Risk 2 debt (PROD03 40% + PROD04 20%), making the minimum mathematically achievable risk score $0.40(1) + 0.60(2) = 1.60$. Our engine detects this exact polyhedral boundary condition and logs an explicit explanation rather than silently breaking the 40% diversification cap.
3. **Contradictory User Inputs (INV06, INV10):** INV10 specifies Risk Tolerance 5.0 alongside `avoid_equity`. Because all Risk 4–5 products in the catalogue are equities, honoring `avoid_equity` restricts the portfolio to debt and hybrid funds (capping achievable risk at 2.06 and return at 8.10%), resulting in an INR 4.94 Lakh shortfall that is clearly explained to the user with a recourse SIP of INR $29,886$/month (+INR $4,886$/month).
4. **Severe Goal Infeasibility & Emergency Buffer Carve-Out (INV08):** INV08 targets INR $50,00,000$ in 36 months with $P =$ INR $2,00,000$, $S =$ INR $20,000$, and only 1.5 months of emergency coverage. Total principal contributed is only INR $9,20,000$. Even at the maximum legal 36-month return (8.28%), Base FV is INR $10,70,319$ (shortfall of INR $39,29,681$). **Smart Wealth** triggers the Emergency Buffer warning (demonstrating an INR $45,000$ liquid reserve in PROD01/PROD03) and computes the exact recourse required: increasing monthly SIP to INR $1,16,389$/month or extending the horizon to 138 months.
5. **Zero Initial Lump Sum & Minimum Buy-In Pruning (INV11):** INV11 has $P =$ INR 0, $S =$ INR $3,000$, and preference `no_lock_in`. PROD02 (minimum investment INR $10,000$, 24m lock-in) is automatically excluded by both the buy-in threshold filter [8] and lock-in filter, and annual product cost (INR 318/yr) is computed on first-year annualized SIP commitments rather than returning a misleading zero cost.

6 Verification, Testing & Possible Future Improvements

- **Automated Verification Suite (`tests/test_engine.py`):** Includes 6 comprehensive unit test suites verifying: (1) non-negative weights summing to $100\% \pm 0.1\%$ and single-product cap $\leq 40\%$; (2) lock-in $\leq$ horizon and horizon risk-bucket limits; (3) weighted risk score $\leq RT + 0.25$ and preference overrides; (4) graceful fallback on missing, negative, zero-capital, or NaN profiles; (5) Section 5 Future Value math including $r = 0$; and (6) Emergency Buffer triggers, What-If shocks, and Softmax Attention normalization.
- **Future Improvements:** (i) Multi-period dynamic programming glide paths using backward Bellman recursion over a discretized wealth grid [5]; (ii) Indian tax-lot harvesting (LTCG/STCG differentiation between equity and debt mutual funds) [10]; and (iii) multi-goal household budget coupling where a shared monthly savings budget is jointly optimized across competing mental accounts [4, 2].

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